

PART-AName of the Student: Bharshita DekaSignature of the Student: Bharshita DekaRoll No.: 230123016Signature of the Invigilator: **INSTRUCTIONS**

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	10	3	08

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Sol) Given set of eqns can be written in matrix form as. $Y = \theta\beta + \epsilon$
 where, $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$, $\epsilon = \text{error}$. y_i 's are observed values of Y .

So our objective function $Q = (Y - \theta\beta)^T(Y - \theta\beta) = \sum \epsilon_i^2$

We need to minimise Q wrt to $\theta = (\theta_1, \theta_2)$

$$Q = Y^T Y - Y^T \theta - \theta^T X^T Y + \theta^T X^T X \theta$$

$$= Y^T Y - 2Y^T \theta + \theta^T X^T X \theta$$

$$Y^T X \theta = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

$$= \theta_1 (y_1 + 2y_2 + 2y_3) + \theta_2 (y_1 - 2y_2 - y_3)$$

$$\theta^T X^T Y = \begin{bmatrix} \theta_1 & \theta_2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= \theta_1 (y_1 + y_2 + 2y_3) + \theta_2 (2y_1 - 2y_2 - y_3)$$

(Since $Y^T X \theta$ is a symmetric matrix $\Rightarrow (Y^T X \theta)^T = \theta^T X^T Y$)

$$= Y^T Y - 2\theta^T X^T Y + \theta^T X^T X \theta = Y^T X \theta$$

$$\Rightarrow Q = Y^T Y - 2Y^T \theta + \theta^T X^T X \theta$$

For Q to be minimum, $\frac{\partial Q}{\partial \theta} = 0$.

$$\Rightarrow \frac{\partial}{\partial \theta} (Y^T Y - 2Y^T \theta + \theta^T X^T X \theta) = 0$$

$$\Rightarrow 0 - 2Y^T + \frac{\partial}{\partial \theta} (\theta^T X^T X \theta) = 0 \quad \text{--- (1)}$$

There is a matrix property, $\frac{\partial}{\partial x} (x^T A x) = (A + A^T)x \Rightarrow 2Ax$ (if A is symmetric)

$$\Rightarrow (\theta^T X^T X \theta)^T = \theta^T X^T X \theta \Rightarrow \text{It is a symmetric matrix}$$

$$\Rightarrow \frac{\partial}{\partial x} (\theta^T X^T X \theta) = 2(X^T X)\theta$$

Put this in (1) $\Rightarrow 2X^T X \theta = 2Y^T$

$$\Rightarrow (X^T X) \theta = Y^T$$

$$\Rightarrow \boxed{\theta = (X^T X)^{-1} Y^T} \rightarrow \text{The LSE}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{27} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \times \frac{1}{9}$$

$$(X^T y) = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{bmatrix}$$

$$\Rightarrow \theta = (X^T X)^{-1} X^T y = \frac{1}{9} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}_{2 \times 2} \times \begin{pmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{pmatrix}_{2 \times 1} = \frac{1}{9} \begin{bmatrix} 3y_1 + 3y_3 \\ 3y_1 - 3y_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} y_1 + y_3 \\ y_1 - y_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} \frac{y_1 + y_3}{3} \\ \frac{y_1 - y_2}{3} \end{bmatrix} \rightarrow \text{LSE}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Sol)

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

$$X_1 = U_1$$

$$X_2 = U_2$$

$$X_3 = U_1 + U_2 = X_1 + X_2$$

var cov matrix of $\mathbf{X} = \begin{bmatrix} \text{Var}(X_1) & 0 & \text{cov}(X_3, X_1) \\ 0 & \text{Var}(X_2) & \text{cov}(X_3, X_2) \\ \text{cov}(X_1, X_3) & \text{cov}(X_2, X_3) & \text{Var}(X_1 + X_2) \end{bmatrix}$

~~$\rho_{mf} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{2\rho_{13}}{\sqrt{2-\rho_{13}}} \\ 0 & \frac{1}{2} & \frac{2\rho_{23}}{\sqrt{2-\rho_{23}}} \\ \frac{2\rho_{13}}{\sqrt{2-\rho_{13}}} & \frac{2\rho_{23}}{\sqrt{2-\rho_{23}}} & 1 \end{bmatrix}$~~

a) $\rho(\mathbf{X}) = \begin{bmatrix} \frac{1}{2} & 0 & \sqrt{2}\rho_1 \\ 0 & \frac{1}{2} & \sqrt{2}\rho_2 \\ \sqrt{2}\rho_1 & \sqrt{2}\rho_2 & 1 \end{bmatrix}$

$$\begin{aligned} \text{Var}(X_1 + X_2) &= \text{Var}(X_1) + \text{Var}(X_2) \\ &\quad - 2\rho_{12} \times \sqrt{\text{Var}(X_1)\text{Var}(X_2)} \\ &= \frac{1}{2} + \frac{1}{2} - 2 \times 0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \rho_1 &= \text{cov}(X_1, X_3) \\ \rho_2 &= \text{cov}(X_2, X_3) \end{aligned}$$

b) $\text{Var}(\Psi_i) = \lambda_i$

$$Y_i = e_i' X_i$$

$$|\mathbf{R} - \lambda \mathbf{I}| = 0$$

$$\Rightarrow \begin{vmatrix} \frac{1}{2} - \lambda & 0 & \sqrt{2}\rho_1 \\ 0 & \frac{1}{2} - \lambda & \sqrt{2}\rho_2 \\ \sqrt{2}\rho_1 & \sqrt{2}\rho_2 & 1 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \frac{1}{2} - \lambda & 0 & \sqrt{2}\rho_1 \\ 0 & \frac{1}{2} - \lambda & \sqrt{2}\rho_2 \\ \sqrt{2}\rho_1 & \sqrt{2}\rho_2 & 1 - \lambda \end{vmatrix} = 0$$

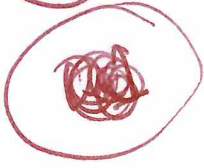
$$\Rightarrow \left(\frac{1}{2} - \lambda\right) \left[\frac{1}{2} - \frac{\lambda}{2} + \lambda^2 \right] + \sqrt{2}\rho_1 \left[\left(\lambda - \frac{1}{2}\right) \sqrt{2}\rho_1 \right] = 0$$

$$\Rightarrow \left(\lambda - \frac{1}{2}\right) \left(\lambda^2 - \frac{3\lambda}{2} + \frac{1}{2} \right) = \sqrt{2}\rho_1^2 \left(\lambda - \frac{1}{2} \right)$$

$$\Rightarrow \left(\lambda - \frac{1}{2}\right) \left[\lambda^2 - \frac{3\lambda}{2} + \frac{1}{2} - 2\rho_1^2 \right] = 0 \Rightarrow \lambda = \frac{1}{2} \text{ (on } \lambda^2 - \frac{3\lambda}{2} + \frac{1}{2} - 2\rho_1^2 = 0)$$

Hence

$$\text{var}(Y_i) = \lambda_i = \frac{1}{2}$$



$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$



3. Let the matrix of distance be

	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) Complete linkage $\Rightarrow d(UV, W) = \max(d_{UV}, d_{WV})$.

$$\begin{array}{c} \Rightarrow (1,3) \\ \Rightarrow (1,3) \end{array} \begin{array}{c} 2 \quad 4 \\ \begin{bmatrix} 0 & & & \\ 2 & 0 & & \\ 4 & 5 & 3 & 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{c} (1,3,4) \\ (1,3,4) \end{array} \begin{array}{c} 2 \\ \begin{bmatrix} 0 & & \\ 3 & 0 \end{bmatrix} \end{array}$$

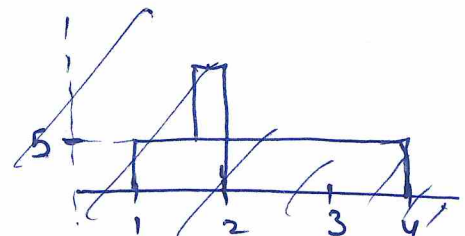
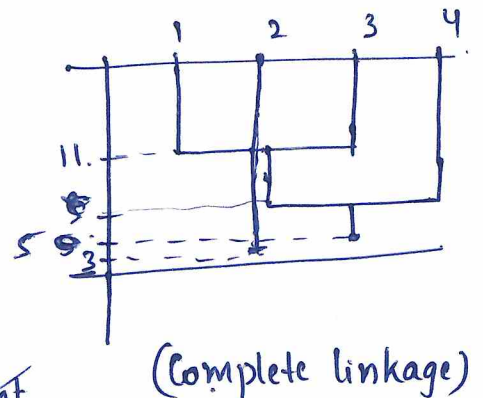
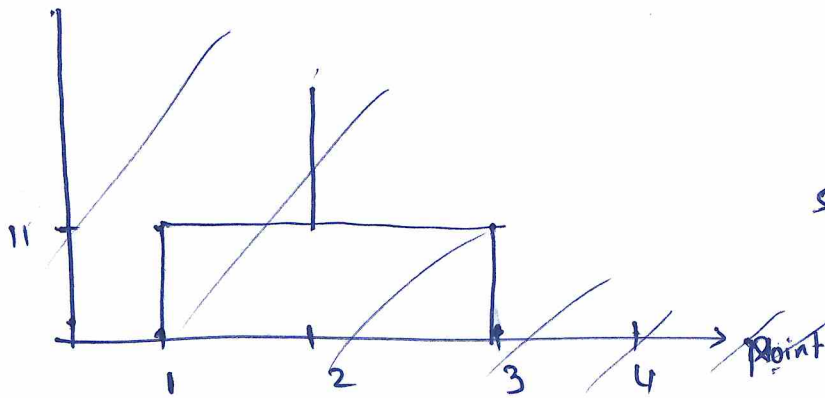
Average linkage $\Rightarrow d(UV, W) = \frac{\sum d_{ik}}{N(UV) \cdot N(W)}$ $\left(\begin{array}{l} d_{ik} = \text{dist. b/t } i^{\text{th}} \text{ item in } UV \text{ \& } j^{\text{th}} \text{ item in } W \\ N_{UV}, N_{W} = \text{Total items in cluster.} \end{array} \right)$

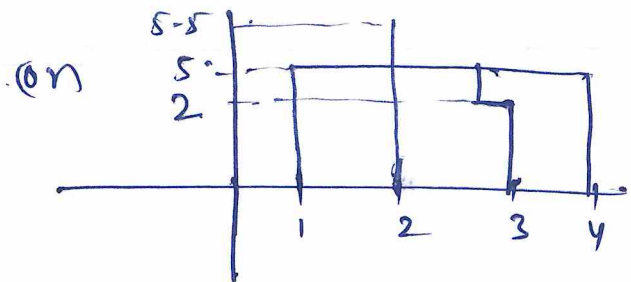
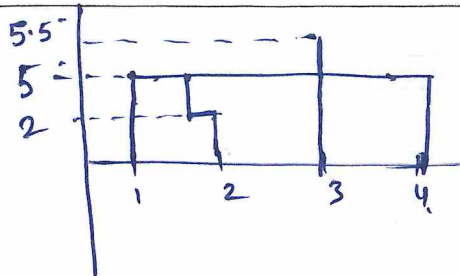
$$\begin{array}{c} \Rightarrow (1,4) \\ \Rightarrow (1,4) \end{array} \begin{array}{c} 2 \quad 3 \\ \begin{bmatrix} 0 & & \\ 2 & 0 & \\ 3 & 7.25 & 2 & 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{c} (1,2,4) \\ (1,2,4) \end{array} \begin{array}{c} 3 \\ \begin{bmatrix} 0 & & \\ 5.5 & 0 \end{bmatrix} \end{array}$$

or

$$\begin{array}{c} (1,3,4) \\ (1,3,4) \end{array} \begin{array}{c} 2 \\ \begin{bmatrix} 0 & & \\ 5.5 & 0 \end{bmatrix} \end{array}$$

ii)





So we can see that ~~the~~ in first method the final separation distance is 3.5 (1, 3, 4 & 2).

whereas in 2nd method it is 5.5.

So we can infer that from both, the ~~2nd~~ method is a better way since more separation/distance means ~~more~~ less similarity.

Hence 1st is a

So we can infer that in avg linkage 2 is more separated from the cluster ~~to~~ that complete linkage.

So acc in avg linkage 2 has lesser similarity w/ wrt cluster in comparison with the complete linkage.

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 ADDITIONAL PAGES FOR ANSWERS
